

	Shri Gurbind Singhji Institute of Engineering and Technology, Nanded Department of BSH&M In-Semester Examination II Academic Year: 2025-26 (Semester – II)		
Class: FY B.Tech.	Subject: Professional Communication	Subject Code: AES102/IIMC110	
Date: 20.11.2025	Time: 04:30 to 05:15 pm	Max. Marks: 20	

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange, Sharing of stationery, calculator etc. not allowed.
- Figures to the right indicate full marks.
- The symbols carry usual meaning.

Q. No.	Attempt the following questions.	Marks	CO	BT
Q.I	Correct the following sentences and rewrite it. (Any ten)	10	CO2	BT2
	1. My sister knit sweater.			
	2. Each of the book racks in the row are pained.			
	3. The astronauts are ready to enter the spacecraft, isn't he?			
	4. Inspiring by Kalpana Chawla, many young girls opt for aerospace engineering.			
	5. Himalayas lies to the north of India.			
	6. Jayant is to weak too work or move about.			
	7. There was group discussion between all the students.			
	8. It will took ages to know the ultimate truth.			
	9. Who did sang a song in Utsav?			
	10. Every man and woman were killed in the war.			
	11. The peoples think of this act.			
Q.II	Convert the following sentences into direct/indirect speech. (Any five)	05	CO3	BT3
	1. Our teacher said, "The sun sets in the west."			
	2. Tony said to his teacher, "Please give me another day to complete my work."			
	3. I said, "You can borrow my pen."			
	4. Captain asked, "Don't allow any outsider in the campus."			
	5. Vinit said to shopkeeper, "What is the price of this tray?"			
	6. Padma congratulated Fancis on his promotion.			
Q.II	Write a paragraph on any one of the topics given.	05	CO4	BT4
	Artificial intelligence and ethics. Or White collard terrorism.			

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To enable students to learn the structure and style of effective sentences.
CO 2	To enable students to express themselves in English effectively.
CO 3	To enhance students' Listening, Speaking, Reading & Writing Skills.
CO 4	To help students to understand nuances of technical writing.

S.G.G.S. Institute of Engineering & Technology, Nanded.

Department of BSHM

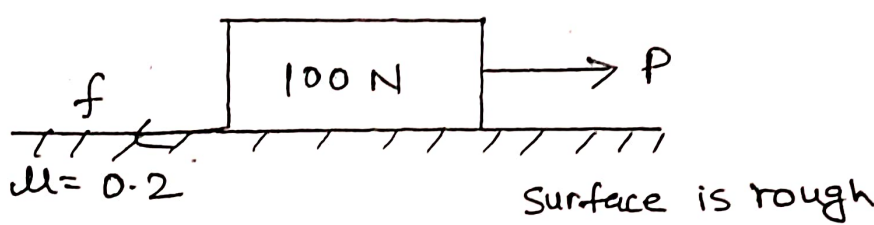
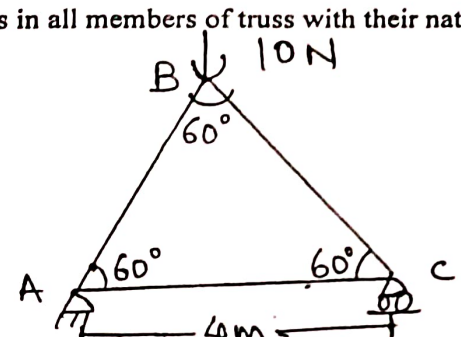
ISE-II EXAMINATION

Date: 21.11.2025
Class: F.Y. Civil

Subject: Engineering Mechanics
Code: ESC 109

Time: 16.30 – 17.15
Marks: 20 (Max)

Note (i) Attempt all questions
(ii) Use of non-programmable calculator is permitted.
(iii) This paper contains 01 page.

Q.No.	Questions	Marks	CO	BT
Q.No.1	Explain following terms 1. Law of lifting machines 2. Levers, Simple axle and wheel	04	CO4	BT2
Q.No.2	Analyse external force (P) acting on the block. 	08	CO2	BT4
Q.No.3	Evaluate forces in all members of truss with their nature. 	08	CO2	BT5

Course Outcome (CO)	Bloom's Taxonomy (BT)
CO1 Determine the resultant force and moment for a given system of forces. CO2 Analyze planar systems to determine the forces in members of trusses, frames and problems related to friction. CO3 Determine the centroid and second moment of area. CO4 Understand the laws and use of simple machines. CO5 Calculate the motion characteristics of a body subjected to a given force system CO6. Apply basic knowledge of mathematics and physics to solve simple real-world problems related to force systems on rigid bodies	BT1-Remember BT2- Understand BT3- Apply BT4- Analyze BT5 – Evaluate BT6 – Create



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded
Department of Production Engineering
Academic Year: 2025-26 (semester I)
In-Semester Examination / Evaluation - II

Class: First Year (Gen.)	Course Name: Engineering Graphics and Design	Course Code: ESC102/106
Date: 19.11.2025	Time: 16:30 to 17:15	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.1	Solve the following questions (Any ONE)	Marks	CO	BT
a	A rectangular plane of edges 40 mm and 70 mm is resting on an edge in the HP. The surface is inclined to the HP such that the top view appears as a square. Draw its projections when the edge resting on the HP is inclined at 30° to the VP.	12	CO1	BT3
b	A plate having the shape of an isosceles triangle has a base 50 mm and altitude 65 mm. It is so placed that in the elevation it is seen as an equilateral triangle of side 50 mm having a side inclined at 45° to the reference line. Draw its plan.	12	CO1	BT3
Q.2	Solve the following questions (Any ONE)			
a	A hexagonal pyramid of base edge 40 mm and altitude 70 mm rests on one of its base edges on the HP with its axis inclined at 30° to the HP and parallel to the V.P. Draw its top and front views.	08	CO1	BT6
b	A square pyramid of base side 40 mm axis length 60 mm has one of its triangular faces in the VP and the axis parallel to the HP. Draw its projections.	08	CO1	BT6

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Draw projections of point, line, plane and solid using the fundamental principles.
CO 2	Read, interpret and draw orthographic projections.
CO 3	Read, interpret and draw isometric projections.
CO 4	Draft various Geometrical Elements used in Engineering Practice using CAD software.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

Academic Year: 2025-26

Semester: I

Examination: ISE-II

Class: First Year

Branch: Basic Science

Date: 22/11/2025

Time: 10:00 - 10:45

Subject: Biology for Engineers

Code: BSC105

Maximum Marks: 20

General Instructions:

- I. This paper contains 1 (One) page. Figures to right indicate full marks.
- II. Answer any four question.
- III. Draw neat & compact sketches wherever necessary.
- IV. Assume data wherever required and state them clearly.

Que.	Question	Marks	CO	BT
1.	Compare and contrast between aerobic and anaerobic respiration.	5	CO2	BT2
2.	Explain Nutrition and classes of Nutrients or food substances.	5	CO2	BT4
3.	Explain an applications of recombinant DNA Technology.	5	CO3	BT3
4.	Explain introduction of the cloning.	5	CO3	BT1
5.	Explain in detail the human digestive system.	5	CO2	BT2

****** ALL THE BEST ******

CO1: Introduction to basics of biology which includes cell, the unit of life, different types of cells, classification of living organisms.

CO2: Understanding what are biomolecules present in a cell, the structures function, and their role in a living organism.

CO3: Application of a certain biomolecules in an industry.

CO4: How biology can be applied in our daily life using different technology for production of medicines to transgenic plants and animals to designing new biological products.

BT1: Remember, **BT2:** Understand, **BT3:** Apply, **BT4:** Analyze, **BT5:** Evaluate, **BT6:** Create.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of BSII&M
Academic Year: 2025-26 (semester I)
In-Semester Examination / Evaluation II

Class: First Year B. Tech.	Course Name: Engineering Thermodynamics and Mechanics	Course Code: PCC-ME 103 ESC 105
Date: 21.11.2025	Time: 16:30 to 17:15	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones are strictly prohibited.
- iii. Non-programmable calculators are allowed.
- iv. Exchange/Sharing of stationery, calculator etc. not allowed.
- v. Figures to right indicate full marks.
- vi. The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	Draw and explain the cone pulley drive along with its applications.	6	CO2	BT2
Q.2	Sketch and explain the compound gear train along with its applications.	6	CO2	BT2
Q.3	With the help of a neat sketch, describe the type of gear arrangement used to transmit power between two shafts that are at right angles and do not intersect	8	CO2	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Apply the thermodynamic laws to analyse the thermodynamic systems.
CO 2	Illustrate various basic parts and transmission systems.
CO 3	Understand the need and steps of machine design.
CO 4	Determine resultant of various force systems.
CO 5	Apply the principles of static and dynamic mechanics to solve problems related to forces, equilibrium, motion, and friction.



**Shri Guru Gobind Singhji Institute of Engineering and Technology,
Vishnupuri, Nanded, Maharashtra, India**

**ISE 2 (Semester-I) 2025-26
Department: First year**

**F.Y. B. Tech
Programming for Problem
Solving ESE103 / ESE107 /
PCC-CS103**

Date: 21/11/2025 Time: 10:00am

Max. Marks : 20

- Note: i. Attempt any 4 questions
ii. Assume suitable data if necessary

Q.N.	Question	Marks	CO	BT
Q.1	Write a C program to accept 'n' random numbers from the user and sort them in ascending order using the Bubble Sort technique. Explain the logic used in the program.	5	CO2	BT4
Q.2	Write a C program to accept a string and a character from the user. The program should count and display the number of times the given character appears in the string. Explain the logic used in the program.	5	CO4	BT4
Q.3	Explain the concept of one-dimensional arrays and two-dimensional arrays in C programming with suitable examples. Also write any two advantages and two limitations of using arrays.	5	CO1	BT1
Q.4	Write a program to get two 3x3 matrices as input and subtract them and store the result in third matrix.	5	CO3	BT3
Q.5	Write a program to check if a number is palindrome or not? Explain logic of the program?	5	CO4	BT2

Course Outcomes (CO)

- CO1. To formulate simple algorithms for arithmetic and logical problems.
CO2. To translate the algorithms to programs (in C language).
CO3. To test and execute the programs and correct syntax and logical errors.
CO4. To implement conditional branching, iteration, and recursion.
CO5. To decompose a problem into functions and synthesize a complete program using divide and conquer approach.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electrical Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester – I)

Class: F.Y (GENERAL)	Subject: Basic Electrical Engineering	SUBJECT CODE- ESC101/105/PCC EL103
Date: 20.11.2025	Time: 10.00 am to 10.45 am	Max. Marks: 20

Note:

- i. *All Questions are compulsory.*
- ii. *Figures to right indicate full marks.*
- iii. *Question paper contains only 02 pages.(See backside of paper)*

Q.N.	Questions	Marks	CO	BT
Q.1	a) State Fleming's Right hand rule and Fleming's Left hand rule b) State Faraday's laws of electromagnetic induction (Both laws)	4	CO-3	BT-1
Q.2	An wooden ring has a mean diameter of 60cm and uniform cross section 9 cm square. It is uniformly wound with a coil of 1800 turns find i) Magnetic field strength produced within coil current of 7.5A. ii) Flux density produced by this current.	4	CO-3	BT-3
Q.3	An iron ring 30 cm in diameter and 20 cm square in cross sectional area is wound with 400 turns. For a flux density of 2Wb/m ² and permeability 1000 find: i) Reluctance of the iron ring ii) Flux in the ring iii) M.M.F. required for iron ring. iv) Exciting current.	4	CO-3	BT-3
Q.4	Draw one complete cycle of the waveform. Calculate the peak value, average value, and RMS value of the AC waveform $v=10\sin\omega t$ Show all steps clearly.	4	CO-4	BT-3
Q.5	Define i) Active power ii) Reactive power iii) Apparent power iv) Complex power	4	CO-4	BT-

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Describe the basic concepts of voltage, current, EMF, energy sources, and fundamental laws like Ohm's and Kirchhoff's in electric circuits.
CO 2	Analyze DC electrical circuits using series-parallel combinations, mesh and nodal analysis, and theorems such as Superposition, Thevenin's, and Norton's.
CO 3	Apply electromagnetic laws to analyse magnetic circuits, induced EMF, and energy storage in magnetic fields.
CO4	Analyze single-phase and three-phase AC circuits using phasor diagrams, and calculate impedance, power factor, and power components.
CO5	Explain the construction and operation of single-phase transformers and basic electrical measuring instruments like PMMC meters and wattmeters.



**SHRI GURU GOBIND SINGHJI INSTITUTE OF ENGINEERING AND
TECHNOLOGY,
DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION
ENGINEERING**

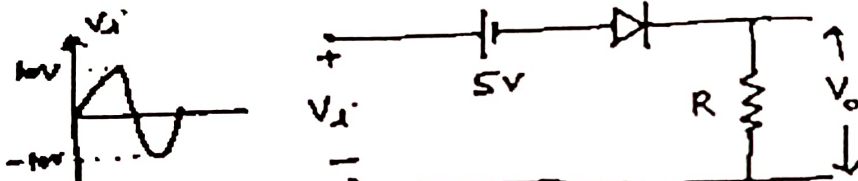
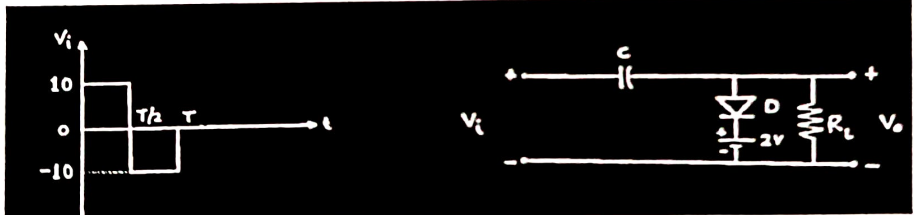
ISE EXAMINATION – II

Academic Year 2025-26 (Semester - I)

Class: FY B. Tech Semester: First	Course: Basic Electronics	Course Code: PCC- EC/EL/IN 102 PCC-EC 103 ESC 104
Date: 19/11/2025	Time: 10:00 AM To 10:45 AM	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- All questions are compulsory.**
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange /sharing of stationery, calculators etc. are not allowed.
- Figures to the right indicate full marks.
- The symbols carry the usual meaning.

Q. No	Questions	Marks	CO	BT
Q.1	Plot the output waveform for the given circuit 	05	CO3	BT5
Q.2	Sketch the output waveform for the given circuit 	05	CO3	BT4
Q.3	Explain common base configuration (CB) with neat diagram and their i/p and o/p characteristics	05	CO2	BT3
Q.4	Write a note on Zener diode	05	CO3	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, **BT2** - Understand, **BT3** - Apply, **BT4** - Analyze, **BT5** - Evaluate, **BT6** - Create

Course Outcomes:

CO1	To introduce some of the basic electronic components and circuits.
CO2	To study the characteristics and working of BJT.
CO3	To explain the applications of components and devices.
CO4	To understand the concept of transistor biasing and its applications.
CO5	To know oscillators and operational amplifiers.



**Shri Guru Gobind Singhji Institute of Engineering and Technology,
Nanded**
Department of BSII&M (Mathematics),
In Semester- Examination-II, Academic Year: 2025-26 (Semester-I)

Class: FY B.Tech.

**Subject: Mathematics-I
(Calculus and linear algebra)**

Subject Code: BSC103

Date: 17/11/2025

Time: 10:00 to 10:45am.

Max. Marks: 20

Note: i) calculators are not allowed.

Q.1	Attempt any four.	Marks	CO	BT
	a i) Find the Maclaurin's series for the function $f(x) = \sin^2 x$. (i.e. Taylor's series about $a = 0$). ii) Obtain the Taylor's series about $a = 0$ for $f(x) = \ln(1 - x^3)$ using geometric series on $(-1, 1)$.	03	CO 1	BT- I
		02		
	b i) Calculate the approximate value of $\sqrt{11}$ by using Taylor's polynomial of order 3. (Final answer in fraction i.e. in p/q form is allowed) ii) Evaluate $\lim_{x \rightarrow \infty} (x - \ln x)$.	03	CO 1	BT- II
		02		
	c i) Using L'Hôpital's rule evaluate $\lim_{x \rightarrow 0^-} \frac{\tan x - x}{x^3}$ ii) Evaluate $\lim_{x \rightarrow \infty} x^{1/x}$	03	CO 2	BT- I
		02		
	d Find row echelon form of the matrix $A = \begin{bmatrix} 0 & 2 & 2 & 1 \\ 3 & -2 & 0 & 1 \\ 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$ and find its rank.	05	CO 5	BT- I
	e Find all the solutions of the following system of simultaneous equations $\begin{aligned} x + 3y + 13z + 3w &= 0 \\ 2x + 3y + 2z - 3w &= 0 \\ 2x + y + 4z - 9w &= 0 \\ 3x + 4y - z - 6w &= 0 \end{aligned}$	05	CO 5	BT- III

[BT-I: Remembering, BT-II: Understanding, BT-III: Applying, BT-IV: Analysing, BT-V: Evaluating, BT-VI: Creating]

CO1: Apply Rolle's Theorem, Mean Value Theorem, and Taylor/Maclaurin series to analyse and approximate functions.

CO2: Evaluate limits involving indeterminate forms using L'Hôpital's rule and solve problems involving maxima, minima, and curve tracing in polar coordinates.

CO3: Solve definite and improper integrals and apply Beta and Gamma functions to evaluate areas and volumes of revolution.

CO4: Solve higher-order ordinary differential equations using methods such as variation of parameters and series solutions, including applications involving Legendre and Bessel functions.

CO5: Analyse and solve systems of linear equations using matrix methods (Gauss-elimination and Gauss-Jordan).

CO6: Compute eigenvalues and eigenvectors of matrices and use Rayleigh's power method to determine dominant eigenpairs and understand orthogonality.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of XXXXXXXXXXXX

Academic Year: 2025-26 (semester I)

In-Semester Examination II / End Semester Examination

Class: First Year

Course Name: Basics of Textile and
Garment Manufacturing

Course Code: PCC-TT102

Date: 17.11.2025

Time: 16:30 to 17:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	Mention and compare different fabric manufacturing methods?	5	CO2	BT4
Q.2	Briefly describe the different steps involved in textile wet processing?	5	CO4	BT1
Q.3	Define technical textile? mention any four areas of technical textiles with suitable examples.	5	CO5	BT2
Q.4	Define the following: a) Warp b) Weft	2	CO1	BT1
b)	Name the basic 3 weaves used in textiles and draw any one?	3	CO3	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Students will learn and understand the basics of textiles and apparels.
CO 2	Students will also learn and understand the manufacturing methods of textiles and apparel.
CO 3	This course will make the students competent to analyze and identify the textiles and garments.
CO 4	This course will enable students to learn practical knowledge of production methods.
CO 5	This course at the highest level may also provide the knowledge of textile and garment industry and their machines.

S.G.G.S. Institute of Engineering & Technology, Nanded.

Department of Civil Engineering

ISE-II EXAMINATION

Subject: Elements of Civil Engineering and Mechanics

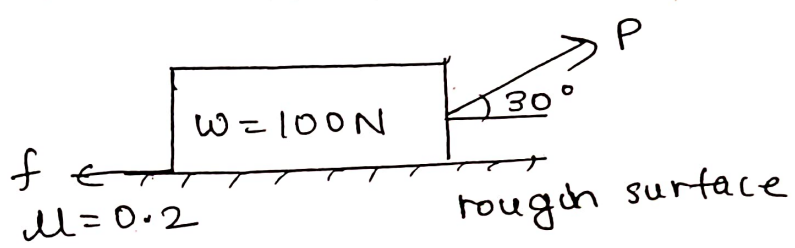
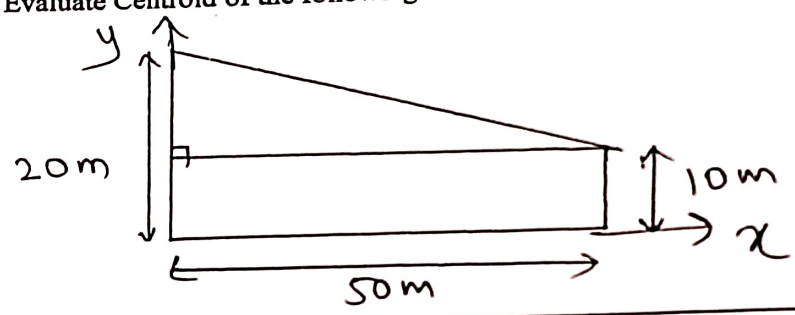
Code: PEC-CE102

Time: 16.30 – 17.15
Marks: 20 (Max)

Note (i) Attempt all questions

(ii) Use of non-programmable calculator is permitted.

(iii) This paper contains 02 page.

Q.No.	Questions	Marks	CO	BT
Q.No.1	Explain following terms 1. Laws of friction. 2. Parallel axis theorem	04	CO4	BT2
Q.No.2	Analyse external force (P) acting on the block. 	08	CO3	BT4
Q.No.3	Evaluate Centroid of the following section. 	08	CO4	BT5

Course Outcome (CO)	Bloom's Taxonomy (BT)
CO1 Know basics of Civil Engineering, its scope of study, knowledge about various fields. CO2 Comprehend the action of Forces, Moments, and other loads on systems of rigid bodies. CO3 Compute the reactive forces and the effects that develop because of the external loads. CO4 Locate the Centroid and compute the Moment of Inertia of regular cross sections. CO5 Express the relationship between the motions of bodies and equipped to pursue studies in allied courses in Mechanics.	BT1-Remember BT2- Understand BT3- Apply BT4- Analyze BT5 – Evaluate BT6 – Create



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of BSH&M
Academic Year: 2025-26 (semester I)
In-Semester Examination / Evaluation II

Class: First Year B. Tech.

Course Name: Elements of
Mechanical and Production
Engineering

Course Code: PCC-ME 102
PCC-PE 102

Date: 17.11.2025

Time: 16:30 to 17:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones are strictly prohibited.
- Non-programmable calculators are allowed.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	Explain any three mechanical properties of materials.	6	CO3	BT2.
Q.2	Derive an expression to determine length of open belt drive.	6	CO4	BT4
Q.3	A prime mover at 300 rpm drives a DC generator at 500 rpm by a belt drive. Diameter of pulley on the output shaft of the prime mover is 60 cm. Assuming a slip of 3 %, determine the diameter of generator pulley, if the belt running over it is 6 mm thick.	8	CO4	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Discuss various thermodynamic properties of mechanical systems.
CO 2	Explain the use and functioning of various energy conversion devices.
CO 3	Discuss the requirements of various mechanical properties for various materials.
CO 4	Identify and recommend power transmission devices for various applications.
CO 5	Illustrate basic concepts of Industrial Management in the fields of different organizations and society and its utilization.
CO6	Determine the application of machining techniques leading to the latest advancements and transmission systems in day-to-day activities.

Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Academic Year: 2025-26 (semester I)
In-Semester Examination / Evaluation II

Class: FY B.Tech.

Course Name: PHYSICS

Course Code: BSC101

Date: 18/11/2025

Time: 10:00 to 10:45am

Max. Marks: 20

Q.N.			Marks	CO	BT
Q.1	a	Which type of lens is used in Newton's Rings experiment? Define a wedge-shaped thin film.	3	CO4	BT1, BT2
	b	If the radius of the 10th dark ring in air is 2.0 mm and, in a liquid, it is 1.8 mm, calculate the refractive index of the liquid.	2	CO2	BT3
Q.2	a	Which material is coated on glass plate A in a Michelson Interferometer? why a partially reflecting coating is applied on glass plate A	3	CO4	BT1
	b	Compare photoelectric effect and Compton effect.	2	CO2	BT4
Q.3	a	In a Michelson Interferometer, 100 fringes shift when air is introduced into a 5 cm cell using light of wavelength 600 nm. Calculate the refractive index of air.	3	CO4	BT3
	b	What experimental evidence confirms the wave nature of electrons?	2	CO2	BT2
Q.4	a	A single-slit diffraction patterns a monochromatic light of wavelength 600 nm passes through a slit of width 0.2 mm. Find the angle for the first minimum.	3	CO4	BT3
	b	Define Rayleigh's criterion of resolution.	2	CO2	BT3

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand the basic concepts of modern, microscopic, and macroscopic theories in physics.
CO 2	Comprehend wave-particle duality and the related experiments.
CO 3	Explain the fundamental experiments and theoretical foundations of quantum mechanics, including Planck's quantum theory and Schrödinger's equation.
CO 4	Analyze wave optics phenomena such as interference, diffraction, and resolution limits, and apply these concepts to optical systems.
CO 5	Understand the fundamental properties of semiconductors and solids, including conductivity, Fermi levels, and the free electron theory.